

Quantum Entanglement

Wire two qubits into a Bell pair, prove the correlation is not classical, and read a CHSH inequality that hardware actually violates.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

In this module a two-qubit ket $|ab\rangle$ lists the first qubit first, as textbooks do, and in CNOT formulas the first qubit is the control. Qiskit prints the same state with the order reversed (lesson 2.5). ρ is a density matrix and Tr_B the partial trace over qubit B .

LESSON 4.1

Two qubits

KEY IDEA

Two qubits have four amplitudes, one per two-bit string. n qubits have 2^n , and most of those states cannot be split into one state per qubit.

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle, \quad \sum_x |\alpha_x|^2 = 1 \quad (4.1.1)$$

the general two-qubit state

Independent qubits combine by the *tensor product*. Each amplitude of the pair is a product of one amplitude from each qubit.

$$(a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle) = a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle \quad (4.1.2)$$

a product state

A product state satisfies $\alpha_{00}\alpha_{11} = a_0b_0a_1b_1 = \alpha_{01}\alpha_{10}$. The converse also holds, which gives a one-line test for whether a two-qubit pure state is entangled.

$$|\psi\rangle \text{ is a product state } \Leftrightarrow \alpha_{00}\alpha_{11} = \alpha_{01}\alpha_{10} \quad (4.1.3)$$

the product test for two qubits

Gates on one qubit of a pair act through the tensor product too: $A \otimes B$ applies A to the first qubit and B to the second, and $A \otimes I$ touches only the first.

$$(A \otimes B)(|a\rangle \otimes |b\rangle) = A|a\rangle \otimes B|b\rangle, \quad \dim(\mathbb{C}^2)^{\otimes n} = 2^n \quad (4.1.4)$$

operators on a register, and its size

QUBITS n	AMPLITUDES 2^n	REAL PARAMETERS $2^{n+1} - 2$	MEMORY FOR A STATEVECTOR
1	2	2	32 bytes
2	4	6	64 bytes
10	1024	2046	16 KB
30	$\approx 1.07 \times 10^9$	$\approx 2.1 \times 10^9$	16 GB

WORKED EXAMPLE · PRODUCT OR ENTANGLED?

$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$: $\alpha_{00}\alpha_{11} = \frac{1}{4} = \alpha_{01}\alpha_{10}$, so it is a product, namely $|+\rangle \otimes |+\rangle$.

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $\alpha_{00}\alpha_{11} = \frac{1}{2}$ but $\alpha_{01}\alpha_{10} = 0$, so no choice of single-qubit states produces it.

LESSON 4.2

CNOT

KEY IDEA

CNOT flips the target when the control is 1. On basis states it is a classical XOR; on a control in superposition it creates entanglement.

$$\text{CNOT } |c\rangle|t\rangle = |c\rangle|t \oplus c\rangle \quad (4.2.1)$$

controlled-NOT: control c , target t

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X \quad (4.2.2)$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

INPUT $ ct\rangle$	OUTPUT	WHAT HAPPENED
$ 00\rangle$	$ 00\rangle$	control 0: nothing
$ 01\rangle$	$ 01\rangle$	control 0: nothing
$ 10\rangle$	$ 11\rangle$	control 1: target flipped
$ 11\rangle$	$ 10\rangle$	control 1: target flipped

CNOT is its own inverse and a Clifford gate. Its behaviour depends on the basis: in the X basis the roles of control and target swap, so information can flow “backwards” from target to control.

$$\text{CNOT}^2 = I, \quad (H \otimes H) \text{CNOT}_{1 \rightarrow 2} (H \otimes H) = \text{CNOT}_{2 \rightarrow 1} \quad (4.2.3)$$

self-inverse, and reversed by Hadamards

The reversal has a useful special case, *phase kickback*. With the target in $|-\rangle$, flipping the target only multiplies it by -1 , and that sign lands on the control instead.

$$\text{CNOT } |c\rangle|-\rangle = (-1)^c |c\rangle|-\rangle \quad (4.2.4)$$

phase kickback: the target is unchanged, the control picks up the sign

WORKED EXAMPLE · KICKBACK ONTO A SUPERPOSITION

Control in $|+\rangle$, target in $|-\rangle$: $\text{CNOT } |+\rangle|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle|-\rangle - |1\rangle|-\rangle) = |-\rangle|-\rangle$.

The target did not change at all, yet the control turned from $|+\rangle$ into $|-\rangle$. Deutsch-Jozsa and Bernstein-Vazirani are built on this.

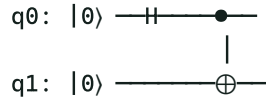
LESSON 4.3

Making a Bell pair

KEY IDEA

A Hadamard puts the control in superposition and a CNOT copies that bit value onto the target. The result is a pair with no individual states and perfectly correlated outcomes.

THE BELL-PAIR CIRCUIT



$$\text{CNOT}(H \otimes I)|00\rangle = \text{CNOT} \frac{|00\rangle + |10\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle \quad (4.3.1)$$

step by step

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}} \quad (4.3.2)$$

the four Bell states: an orthonormal basis of two qubits

The same circuit makes all four, depending on the input bits. The first input bit chooses the sign, the second chooses whether the two outputs agree.

$$|\beta_{xy}\rangle = \text{CNOT}(H \otimes I)|xy\rangle = \frac{|0y\rangle + (-1)^x|1\bar{y}\rangle}{\sqrt{2}} \quad (4.3.3)$$

$$\beta_{00} = \Phi^+, \beta_{10} = \Phi^-, \beta_{01} = \Psi^+, \beta_{11} = \Psi^-$$

$|\Phi^+\rangle$ fails the product test of lesson 4.1, since $\alpha_{00}\alpha_{11} = \frac{1}{2} \neq 0 = \alpha_{01}\alpha_{10}$. Measured in the Z basis it gives 00 or 11, each half the time, never 01 or 10. The correlation also survives a change of basis, which a pair of pre-agreed classical bits cannot manage.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|++\rangle + |--\rangle}{\sqrt{2}} \quad (4.3.4)$$

perfectly correlated in the Z basis and in the X basis

WORKED EXAMPLE · CHECKING THE X-BASIS FORM

$$|++\rangle + |--\rangle = \frac{1}{2}[(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)(|0\rangle - |1\rangle)] = \frac{1}{2}[2|00\rangle + 2|11\rangle] = |00\rangle + |11\rangle.$$

The cross terms $|01\rangle$ and $|10\rangle$ cancel.

LESSON 4.4

No state of your own

KEY IDEA

Half of an entangled pair has no pure state. Its best description is a density matrix, and for a Bell pair that matrix is the maximally mixed one, with a Bloch vector of length zero.

$$\rho_A = \text{Tr}_B |\psi\rangle\langle\psi| \quad (4.4.1)$$

the reduced state of qubit A : what every measurement on A alone can see

$$\rho = \frac{1}{2} (I + r_x X + r_y Y + r_z Z), \quad |\vec{r}| \leq 1 \quad (4.4.2)$$

any one-qubit state, pure ($|\vec{r}| = 1$) or mixed ($|\vec{r}| < 1$)

For $|\Phi^+\rangle$ the partial trace gives $\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = I/2$, so $\vec{r} = 0$. Every measurement on A alone is a fair coin, in every basis. The information is in the correlations, not in either qubit.

$$\text{Tr}(\rho^2) = \frac{1 + |\vec{r}|^2}{2} \quad (4.4.3)$$

purity: 1 for a pure state, $\frac{1}{2}$ for the maximally mixed one

For a pure state of two qubits, entanglement and the mixedness of each half are the same thing. It can be quantified by the *concurrence* C , which is 0 for product states and 1 for Bell states, and ties directly to the Bloch vector length.

$$C = 2|\alpha_{00}\alpha_{11} - \alpha_{01}\alpha_{10}|, \quad |\vec{r}_A|^2 = |\vec{r}_B|^2 = 1 - C^2 \quad (4.4.4)$$

concurrence, and the shrinking Bloch arrow

$$|\psi\rangle = \sum_i \lambda_i |a_i\rangle |b_i\rangle, \quad S = - \sum_i \lambda_i^2 \log_2 \lambda_i^2 \quad (4.4.5)$$

Schmidt form and entanglement entropy: $S = 1$ ebit for a Bell pair

WORKED EXAMPLE · A PARTLY ENTANGLED PAIR

$$|\psi\rangle = \cos \gamma |00\rangle + \sin \gamma |11\rangle.$$

$$C = 2|\cos \gamma \sin \gamma| = |\sin 2\gamma| \text{ and } \rho_A = \text{diag}(\cos^2 \gamma, \sin^2 \gamma), \text{ so } \vec{r}_A = (0, 0, \cos 2\gamma) \text{ and } |\vec{r}_A|^2 = \cos^2 2\gamma = 1 - C^2.$$

At $\gamma = \pi/8$ both numbers are $1/\sqrt{2} \approx 0.71$: the pair is partly entangled, and each arrow is shorter than a pure state's but still points somewhere.

LESSON 4.5

Measuring one half

KEY IDEA

Measuring one qubit of a Bell pair fixes the other's outcome in that basis. But the other qubit's own statistics, taken without the result, do not change at all.

$$(|a\rangle\langle a| \otimes I) |\Phi^+\rangle = \frac{1}{\sqrt{2}} |a\rangle\langle a| \Rightarrow P(a) = \frac{1}{2}, \text{ state afterwards } |aa\rangle \quad (4.5.1)$$

measuring qubit A of $|\Phi^+\rangle$ and getting a

After the measurement the pair is a product state: C drops from 1 to 0, and both Bloch vectors jump from the origin to a pole. Measuring in any other basis does the same with that basis's states, up to a complex conjugate.

$$(\langle u| \otimes I) |\Phi^+\rangle = \frac{1}{\sqrt{2}} |u^*\rangle \quad (4.5.2)$$

Alice finds $|u\rangle$; Bob's qubit is left in $|u^*\rangle$

The outcome on A is random, so Bob, who has not been told it, holds a 50/50 mixture of the two possible results. That mixture is exactly the reduced state he held before anything was measured.

$$\rho_B = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| = \frac{I}{2} \quad \text{before and after A is measured} \quad (4.5.3)$$

B's own statistics cannot tell whether A was measured

	BEFORE A IS MEASURED	A MEASURED, RESULT UNKNOWN TO B	A MEASURED, RESULT TOLD TO B
State of the pair	$ \Phi^+\rangle$	mixture of $ 00\rangle$ and $ 11\rangle$	$ aa\rangle$
ρ_B	$I/2$	$I/2$	$ a\rangle\langle a $
Concurrence	1	0	0

WORKED EXAMPLE · DIFFERENT BASES, SAME CORRELATION

Alice measures in the X basis and gets $-$. Using the X-basis form of lesson 4.3,

$|\Phi^+\rangle = (|++\rangle + |--\rangle)/\sqrt{2}$, so Bob's qubit is now $|-\rangle$, and an X measurement on his side gives $-$ with certainty. A Z measurement on his side is still a fair coin.

LESSON 4.6

Correlation is not communication

KEY IDEA

Entangled outcomes are correlated more strongly than any pre-agreed classical strategy allows, and the CHSH inequality measures by how much. The correlation still carries no message.

Alice chooses between two measurement settings a and a' , and Bob between b and b' . Each outcome is ± 1 , and $E(a, b)$ is the average of the product of their outcomes.

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b') \quad (4.6.1)$$

the CHSH combination

$$|S| \leq 2 \text{ (any local hidden-variable model),} \quad |S| \leq 2\sqrt{2} \approx 2.83 \text{ (quantum mechanics)} \quad (4.6.2)$$

Bell's bound and Tsirelson's bound

The classical bound follows because, if every outcome is fixed in advance, $S = a(b - b') + a'(b + b')$ with each value ± 1 : one bracket is 0 and the other ± 2 . Quantum pairs do better. For $|\Phi^+\rangle$ measured along directions in the x - z plane at angles θ , the correlation is a cosine.

$$E(\theta_a, \theta_b) = \cos(\theta_a - \theta_b) \quad \text{for } |\Phi^+\rangle, \text{ observable } \cos \theta Z + \sin \theta X \quad (4.6.3)$$

Bell-pair correlations

WORKED EXAMPLE · REACHING 2√2

Choose $a = 0, a' = \frac{\pi}{2}, b = \frac{\pi}{4}, b' = \frac{3\pi}{4}$.

$$E(a, b) = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}, E(a, b') = \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, E(a', b) = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}, E(a', b') = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

$$S = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = 2\sqrt{2}, \text{ beyond anything classical.}$$

None of this sends information. Whatever Alice chooses to measure, Bob's averaged state is the same, so his statistics alone reveal nothing about her choice. The correlation only appears when the two lists of results are brought together, by an ordinary classical channel.

$$\sum_a p(a) \rho_{B|a} = \text{Tr}_A |\Phi^+\rangle \langle \Phi^+| = \frac{I}{2} \quad \text{for every choice of Alice's basis} \quad (4.6.4)$$

no-signalling

Loophole-free Bell tests since 2015 have measured $S > 2$ with the detection and locality loopholes closed at once. Superdense coding and teleportation, in the algorithms section, are what this non-classical correlation is used for.

MODULE 4 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

the general two-qubit state	$ \psi\rangle = \alpha_{00} 00\rangle + \alpha_{01} 01\rangle + \alpha_{10} 10\rangle + \alpha_{11} 11\rangle, \quad \sum_x \alpha_x ^2 = 1$	(4.1.1)
a product state	$(a_0 0\rangle + a_1 1\rangle) \otimes (b_0 0\rangle + b_1 1\rangle) = a_0b_0 00\rangle + a_0b_1 01\rangle + a_1b_0 10\rangle + a_1b_1 11\rangle$	(4.1.2)
the product test for two qubits	$ \psi\rangle$ is a product state $\Leftrightarrow \alpha_{00}\alpha_{11} = \alpha_{01}\alpha_{10}$	(4.1.3)
controlled-NOT: control c , target t	$\text{CNOT } c\rangle t\rangle = c\rangle t \oplus c\rangle$	(4.2.1)
phase kickback: the target is unchanged, the control picks up the sign	$\text{CNOT } c\rangle -\rangle = (-1)^c c\rangle -\rangle$	(4.2.4)
step by step	$\text{CNOT } (H \otimes I) 00\rangle = \text{CNOT } \frac{ 00\rangle + 10\rangle}{\sqrt{2}} = \frac{ 00\rangle + 11\rangle}{\sqrt{2}} = \Phi^+\rangle$	(4.3.1)
the four Bell states: an orthonormal basis of two qubits	$ \Phi^\pm\rangle = \frac{ 00\rangle \pm 11\rangle}{\sqrt{2}}, \quad \Psi^\pm\rangle = \frac{ 01\rangle \pm 10\rangle}{\sqrt{2}}$	(4.3.2)
perfectly correlated in the Z basis and in the X basis	$ \Phi^+\rangle = \frac{ 00\rangle + 11\rangle}{\sqrt{2}} = \frac{ ++\rangle + --\rangle}{\sqrt{2}}$	(4.3.4)
the reduced state of qubit A : what every measurement on A alone can see	$\rho_A = \text{Tr}_B \psi\rangle\langle\psi $	(4.4.1)
any one-qubit state, pure ($ \vec{r} = 1$) or mixed ($ \vec{r} < 1$)	$\rho = \frac{1}{2} (I + r_x X + r_y Y + r_z Z), \quad \vec{r} \leq 1$	(4.4.2)
concurrence, and the shrinking Bloch arrow	$C = 2 \alpha_{00}\alpha_{11} - \alpha_{01}\alpha_{10} , \quad \vec{r}_A ^2 = \vec{r}_B ^2 = 1 - C^2$	(4.4.4)
measuring qubit A of $ \Phi^+\rangle$ and getting a	$(a\rangle\langle a \otimes I) \Phi^+\rangle = \frac{1}{\sqrt{2}} a\rangle a\rangle \Rightarrow P(a) = \frac{1}{2}, \text{ state afterwards } aa\rangle$	(4.5.1)
B 's own statistics cannot tell whether A was measured	$\rho_B = \frac{1}{2} 0\rangle\langle 0 + \frac{1}{2} 1\rangle\langle 1 = \frac{I}{2} \text{ before and after } A \text{ is measured}$	(4.5.3)
the CHSH combination	$S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$	(4.6.1)
Bell's bound and Tsirelson's bound	$ S \leq 2$ (any local hidden-variable model), $ S \leq 2\sqrt{2} \approx 2.83$ (quantum mechanics)	(4.6.2)
no-signalling	$\sum_a p(a) \rho_{B a} = \text{Tr}_A \Phi^+\rangle\langle\Phi^+ = \frac{I}{2} \text{ for every choice of Alice's basis}$	(4.6.4)